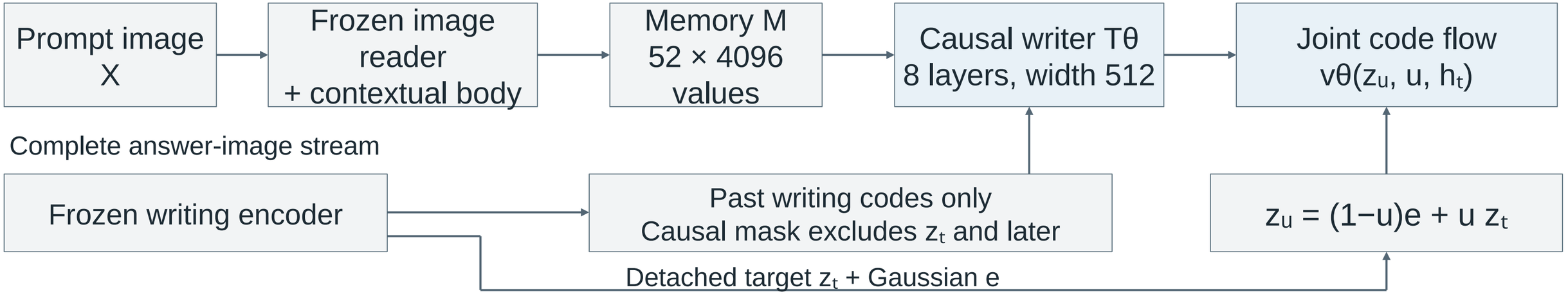
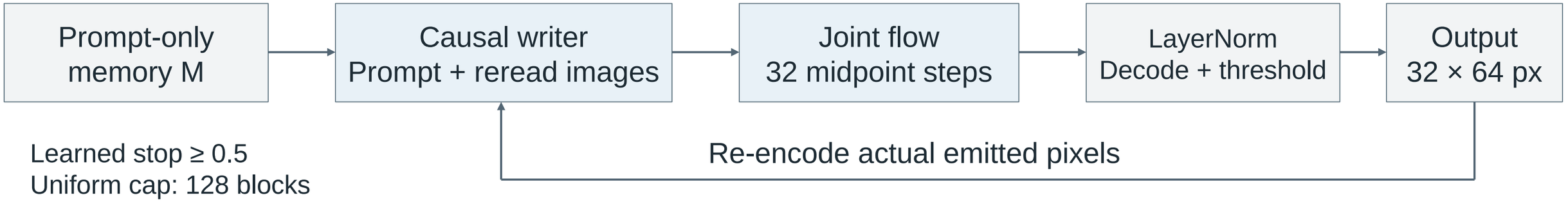
Causal image writing with frozen visual semantics

a Training | complete answer images supervise continuous writing



$$L = \text{mean } ||v\theta - (z_t - e)||^2 + \text{balanced stop loss}$$

b Inference | every generated block becomes physical visual history



38.65M trainable writer. External reader pretraining retained. Completed Chinese audit: 0/128 exact; 29.386% foreground F1.